

ware components such as the camera, gyro and proximity sensor can be played around with and optimised, in theory.

Easy-to-use native tools for reusing code in C++ and Java across operating systems and platforms, which seamlessly integrate with popular Integrated Development Environments (IDEs) such as Eclipse, make development quicker and enable the delivery of higher performance – and thus, more responsive – apps. All the popular tools and technology provided in the suite are completely future-proof to any updates. The Intel® INDE tool provides useful documentation and support with in-hub alerts for new tools as well. This development suite has tools like Intel® XDK and Intel® HAXM, which can be used to compile code and port applications to Java-based Android as well as Microsoft Windows targets. However, the optimisations provided are specific to mobiles powered by ARM architecture and best for Intel® architecture-based devices running Android 4.3 and above, and Microsoft Windows 7/8.1. Additional tools for host and target support have been announced to be launched this year.

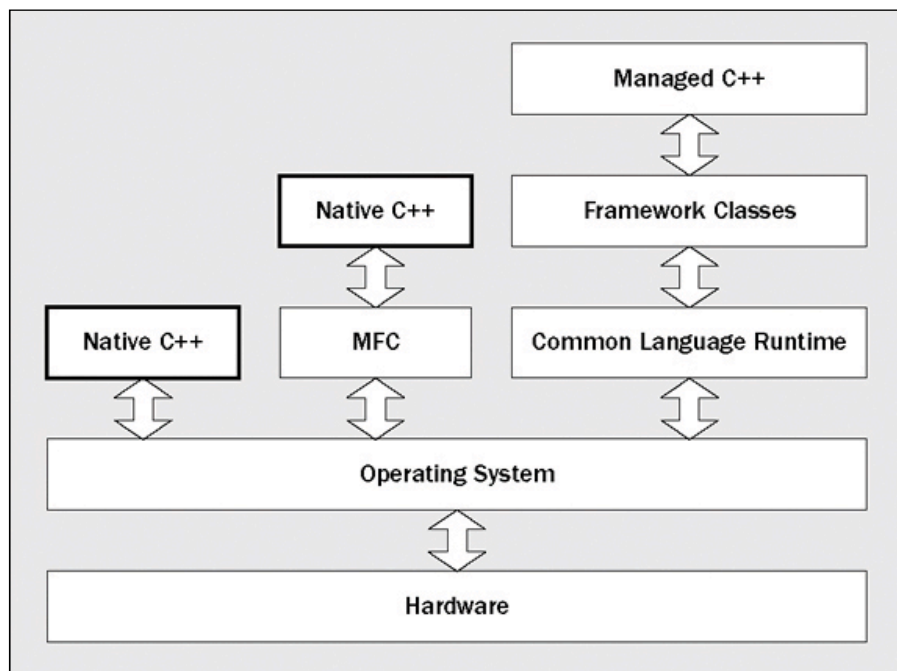
Intel® INDE offers easy-to-use solutions to speed up development for existing and new-to-Android developers using C++ or Java. From quick, professional-grade implementations of media video conferencing, to real-time analysis and debugging of graphics, Intel® INDE offers performance-oriented solutions. This facet of Intel® INDE has been emphasised by its additional ability to compile legacy/native code compared to other IDEs for Android target development. This is essential to perform hardware-specific optimisations with your software. Applications that were made for different architectures (https://en.wikipedia.org/wiki/Addressing_mode) can now be ported to Intel® Atom-based processor devices. For example, if specific hardware architecture is being targeted, it becomes easier for the software developer to code in optimisations, as s/he knows how and where to address the variables without the need for any translations or buffers, thereby easing the load on the hardware.

Intel® INDE runs on 64-bit Microsoft Windows 7/8.1 host systems with a minimum of 4GB RAM and 6GB worth of free space. However, to be able to use Intel® HAXM (Hardware Accelerated Execution Manager), the system must have Intel® Virtualization Technology enabled in BIOS.

Intel® INDE provides access to the following:

- Intel® Tools:
- Intel® INDE Media Pack for Android*
- Intel® Threading Building Blocks
- Compute Code Builder beta
- Intel® C++ Compiler for Android*
- System Analyzer: Part of Intel® Graphics Performance Analyzers (Intel® GPA)
- Platform Analyzer: Part of Intel® GPA
- Frame Analyzer: Part of Intel® GPA

installing only those tools that are required, making the environment highly customisable. There is Eclipse (through the Android SDK for Java development), vs-android plug-in for Microsoft Visual Studio IDE support and Apache Ant for command line development. Other tools include Android NDK for C++ development, Android Design for style library integration and Intel® Hardware Accelerated Execution Manager



Native C++ is used to code the binaries and tools provided by Intel INDE. Optimizations become more efficient as the language gets closer to the hardware.

- Intel® Frame Debugger beta
- Environment Setup: Includes an optional installation of Intel®HAXM

Third-party Tools (Optional components/installations that are part of Environment Setup):

- Google Android* SDK
- Android* NDK
- Android* Design
- Apache Ant*
- vs-android plug-in for Microsoft* Visual Studio*

OVERVIEW OF INTEL® INDE

- **Environment Setup:** Formerly made popular under the codename 'Beacon Mountain', the Environment Setup tool in Intel® INDE quickstarts development by installing an IDE and integrating popular tools. The interface provides the option of

(HAXM) for Android (among others) app emulation in a host environment.

- **Media Pack for Android:** Developers can create and add visually compelling video and audio extensions that are native to the hardware of the mobile. Applications can be developed with ease by using the media component, available through Intel® INDE. The Media Pack for Android* enables visually powerful and easily integrated top video and audio use cases (which determine the preferred bitrates for both), delivering professional quality solutions for camera, screen capturing, video editing, video streaming and audio fingerprinting. Source code and examples are given to help advanced programmers further customise their video transcode pipeline. All libraries are optimised for x86 and ARM* based Android* devices running Jelly Bean 4.3 & above.